FixIt Now: Implementation Plan

Version 1.0 · 16 October 2026 · Companion to the FixIt Now SRS

1. Goal and Approach

Build FixIt Now in vertical slices, so a working end-to-end flow exists early and each later phase adds to it. The first working target is: customer creates a request, a partner accepts it, and the customer sees live status ticks. Subscriptions, KYC and admin features are layered on after that core loop works.

2. Technology Stack

Layer	Choice	Why
Frontend	React with React Router	Component reuse across three portals
Styling	Tailwind CSS	Fast, consistent, responsive UI
Backend	Node.js with Express	Simple REST API, same language as frontend
Database	PostgreSQL	Relational data: users, requests, subscriptions
Real-time	Socket.IO	Status ticks, partner movement, cancellations
Auth	JWT with bcrypt password hashing	Stateless, role-based access
Uploads	Multer	KYC PDFs and photos
OTP	Nodemailer (email) or console fallback	Forgot password and service OTP
Geocoding	OpenStreetMap Nominatim with cache	Free, street-level accuracy
Charts	Chart.js or Recharts	Admin analytics
Map	Custom SVG canvas	Simple partner movement and ETA

3. Architecture Overview

The React app talks to the Express API over HTTPS for data and to Socket.IO over WebSocket for live updates. The API contains separate modules: auth, customers, partners, requests, matching, subscriptions, KYC, payments, ratings, notifications and admin. PostgreSQL stores all data; uploaded files are stored on disk with their paths in the database.

3.1 Suggested folder structure

- `client/`: `pages/` (landing, customer, partner, admin), `components/`, `services/` (API and socket clients), `context/` (auth)
- `server/`: `routes/`, `controllers/`, `models/`, `middleware/` (auth, role checks, upload), `services/` (matching, geocoding, simulation, subscription checks), `sockets/`
- `db/`: schema and seed scripts

3.2 Key real-time events

- `request:new` to a partner, `request:accepted`, `request:declined`, `request:timeout`
- `status:update` to the customer for each job step
- `partner:location` for simulated movement and ETA
- `request:cancelled` to the partner

4. Phases

Durations assume one to two developers and are estimates to adjust to your deadline.

Phase 1: Foundation (about 3 days)

- Set up repositories, project structure, database and environment config.
- Create the database schema for all entities in the SRS and a seed script (admin account, services, plans, demo partners).
- Build the common layout, navbar and routing.

Done when: the app runs, connects to the database and the admin account exists.

Phase 2: Landing page (about 2 days)

- Build all landing sections: hero, services, why FixIt Now, About us, How it works, FAQ, partner call-to-action and footer.
- Load services from the database.

Done when: the landing page is complete and responsive.

Phase 3: Authentication (about 3 days)

- Customer and partner registration with validation (password of at least 6 characters and one special character).
- Login, JWT sessions and role-based route guards.
- Forgot password with OTP; separate admin login at `/admin/login`.

Done when: all three roles can log in and are kept out of each other's routes.

Phase 4: Partner subscription and KYC (about 4 days)

- Plan selection after registration (quarterly, half-yearly, yearly) with simulated payment.
- Subscription records with start and end dates, expiry lock and warnings.
- KYC upload (PDFs and passport photo) with pending, verified and rejected states.
- Partner dashboard gating: "Complete KYC verification" until verified.

Done when: a partner can register, subscribe, submit KYC and see the right dashboard state.

Phase 5: Admin core (about 3 days)

- KYC review screen: approve or reject.
- Plan price management (applies to new subscriptions only) and service category management.
- Read-only lists of customers, partners, requests, subscriptions, ratings and the activity log.

Done when: admin can verify a partner so the partner can start receiving requests.

Phase 6: Request creation and matching (about 5 days)

- Customer dashboard, address saving and geocoding with caching and fallbacks.
- Create-request flow with service, problem, description, optional photo and price estimate.
- Matching service: filter online, verified, subscribed partners of the right service, compute haversine distance, keep those within 2 km, sort nearest first.
- Offer to the nearest partner with a countdown; on decline or timeout, offer to the next. If none remain, return "All partners busy, try again later".
- Customer searching screen and partner incoming-request popup.

Done when: a request reaches the right partner, and the busy case works.

Phase 7: Live job flow and tracking (about 5 days)

- Partner status buttons: Accept, Start trip, Arrived, Start service, Complete service, Payment received.
- Customer tracking screen with six green ticks updated through Socket.IO.
- Simulated partner movement along a straight line with ETA countdown on the custom SVG map.
- Service OTP check before starting the service.
- Customer cancellation with reason and instant partner notification; the partner can also be freed for new jobs.

Done when: a full job runs from request to payment with live ticks on the customer side.

Phase 8: Payment, ratings, history and earnings (about 3 days)

- Simulated payment states: proceed, waiting, successful, plus receipts.
- Rating of the partner after service.
- Request history for both sides, partner earnings with breakdown, ratings view.
- Notifications and support page.

Done when: completed and cancelled jobs appear in history and earnings update after payment.

Phase 9: Admin analytics (about 3 days)

- Charts for requests over time, top services, completion versus cancellation, partner and customer growth, subscription revenue, peak hours and top-rated partners.
- Live users view.

Done when: every analytics item from the SRS is shown with real data.

Phase 10: Polish, testing and demo prep (about 4 days)

- Empty, loading and error states; mobile responsiveness.
- End-to-end testing of all flows; security review of routes and uploads.
- Seed demo data and rehearse the demo script.

Done when: the demo runs cleanly start to finish.

5. Overall Timeline

Roughly 35 days of work in total, or about 5 to 6 weeks. The critical path is Phases 3, 4, 5, 6 and 7, because matching and live tracking cannot be demonstrated until a verified, subscribed partner exists.

6. Testing Strategy

- **Unit tests**: distance calculation, eligibility filter, password rules, subscription expiry, plan price handling.
- **API tests**: auth, role guards, request lifecycle, KYC and admin actions.
- **Manual end-to-end scenarios**: see the demo checklist below.
- **Real-time checks**: two browsers (customer and partner) to confirm tick updates, cancellation and movement.

7. Demo Checklist

1. Landing page walkthrough.
2. Register a partner, subscribe, submit KYC; admin approves.
3. Register a customer and create a request with a real street and area address.
4. Partner within 2 km accepts; walk through all status buttons while the customer sees live ticks, movement and ETA.
5. Service OTP, payment, rating, receipt and history.
6. Customer cancels a second request; partner is notified.
7. Request from an address more than 2 km away shows "All partners busy".
8. Admin views analytics and changes a plan price; show that existing partners keep what they paid.

8. Risks and Mitigations

Risk	Mitigation
Geocoding fails or is slow	Cache results; fall back to area name, then simulated coordinates

Risk	Mitigation
Real-time updates drop during demo	Reconnect logic and a quick refresh fallback on the tracking screen
No eligible partner during demo	Seed demo partners at real nearby areas and keep them online
Scope too large for the deadline	Build core loop first; treat extras (photo upload, receipts, support page) as last
KYC uploads misuse	File type and size limits, files stored outside the public path

9. Open Items to Confirm

- Offer timeout length per partner (suggested 30 seconds, configurable).
- Demo city and areas to seed partners in.
- Whether OTPs are emailed or shown on screen for the demo.
- Your submission deadline, so the phase durations can be adjusted.